

GEOLOGICAL ASSESSMENT & EXPANSION REPORT (2026)

Mine Block: Gevra OpenCast Project (Expansion to 70 MTPA)

Coalfield: Korba Coalfield, Mand-Raigarh Basin

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1. EXECUTIVE SUMMARY

Gevra OpenCast Project (OCP), operated by South Eastern Coalfields Limited (SECL), represents one of the largest sovereign energy assets in India. This technical dossier consolidates verified borehole lithology logs, seam continuity models, and proved coal reserve calculations covering the Seam V, VI, and VII combined block.

KEY CERTIFIED METRICS:

- Total Proved Coal Reserves: 425.80 Million Tonnes (MT) in Seam V/VI/VII.
- Average Seam Thickness: 18.4 meters (cumulative workable coal thickness).
- Overburden Stripping Ratio: 2.14 m³/tonne.
- Dominant Coal Grade: Grade G11 to G13 (Gross Calorific Value: 4,300 - 4,900 kcal/kg).
- Annual Targeted Production: 70.00 MTPA with fully mechanized surface miners.

2. GEOLOGICAL & STRATIGRAPHIC SETTING

The project area is located in the South-Central sector of Korba Coalfield. The sedimentary strata belong to the Barakar Formation of Lower Gondwana age. Major structural faults show throw ranging from 5m to 25m with strike N50W-S50E. The coal seams exhibit remarkable lateral continuity with gentle southerly dip (3° to 5°).

GEVRA OCP · SEAM-WISE PROVED RESERVES & METRICS

3. PROVED RESERVES & QUALITY METRICS BREAKDOWN

The table below details proved in-situ coal reserves, average thickness, stripping ratio, and ash percentages derived from 48 deep exploration boreholes drilled by CMPDI.

Coal Seam Block	Thickness (m)	Proved MT	Grade / GCV	Stripping Ratio (m³/t)
Seam V Upper	12.4 m	145.20 MT	G11 (4,550 kcal/kg)	2.28 m³/t
Seam VI Lower	18.2 m	180.40 MT	G12 (4,380 kcal/kg)	2.05 m³/t
Seam VII Bottom	7.8 m	100.20 MT	G13 (4,250 kcal/kg)	2.10 m³/t
CUMULATIVE TOTAL	18.4 m (Avg)	425.80 MT	G11 - G13 (Avg 4,650)	2.14 m³/t (Overall)

4. BOREHOLE VALIDATION & CONFIDENCE AUDIT

- Primary Borehole Reference: SB-42, BH-704, and Core Log BH-812.
- Specific Gravity of Coal: 1.48 to 1.54 g/cm³.
- Equilibrated Moisture: 6.8% at 60% Relative Humidity and 40°C.
- Volatile Matter (dry mineral matter free basis): 28.5%.
- Total Overburden to be Handled: 911.21 Million Cubic Meters.
- Certified by: CMPDI Exploration Geologist & CIL Reserves Authority.